

Evolving Disease Progression Networks through Incremental Refinement

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Abstract

Healthcare recommender systems are increasingly used for personalized diagnosis and treatment planning. Yet existing approaches fail to capture the evolving nature of clinical data. In particular, two challenges remain insufficiently addressed: first, the need for meaningful temporal prognostication that reflects how diseases progress over time; and second, the ability to adapt to incoming patient data without requiring costly retraining of the underlying inference model. To address these limitations, we propose a dynamic framework for constructing disease progression networks that evolve in near real-time. We construct a directed disease network from co-occurrence data using supervised learning to capture probabilistic influence relationships among diseases. After sparsifying the network to retain informative edges, we introduce and apply an iterative refinement algorithm that updates influence weights from streaming patient trajectories, eliminating the need for retraining. Theoretically, the adjustment mechanism is a stable stochastic approximation process: the bias in edge estimates decays geometrically to zero, ensuring asymptotic consistency, while the variance remains bounded in steady state. Evaluations on the MIMIC-III dataset show that International Classification of Diseases, 9th Revision (ICD9) codes exhibit partial alignment with data-driven disease clusters, revealing that clinical classification does not denote co-occurrence. Our approach achieves strong agreement with ground-truth networks (Kendall’s $\tau \approx 0.8$, Pearson’s correlation coefficient ≈ 0.7) while processing incoming batches of patient data in under one second. It also exhibits robustness by achieving high concordance when applied to a conditional probability-based disease network inference baseline. Validation on quasi-patient trajectories shows that the framework recovers interpretable disease progressions across systems, including cascading failure pathways, cross-system interactions, and chronic cyclic disease dynamics, uncovering rich temporal prognostic structure.

1 Introduction

The modern healthcare landscape is increasingly driven by vast, heterogeneous biomedical data, from electronic health records (EHRs) and laboratory results to genomic profiles and wearable sensor streams, all of which promise to improve predictive clinical insights. Integrating these “big data” is explicitly aimed at enhancing care quality and efficiency, while lowering costs and errors [1]. Recommender systems have emerged as a key tool for harnessing this data deluge in personalized medicine. A healthcare recommender system (HRS) automatically analyzes a patient’s unique profile and suggests highly relevant items, namely, likely diagnoses, tailored treatment options, or preventive care strategies [2]. By filtering vast medical information and highlighting what is most relevant for an individual, HRSs can reduce information overload and direct clinicians and patients toward the most pertinent evidence. In doing so, these systems can support more precise risk stratification and decision-making: by matching patient-specific data to known patterns, they help predict which patients might benefit from particular interventions or monitoring, improving predictive outcomes and personalized interventions [3].

Specifically, disease-disease interactions via network-based models can enhance such predictive recommender systems [4]. Network approaches model diseases (or patients) as nodes connected by links that represent comorbidity or progression relationships. Recent advances show that constructing “disease networks”, capturing statistically significant co-occurrence or transition rates between conditions, provides a powerful way to encapsulate complex multimorbidity patterns. For example, large-scale analyses of administrative health data have used correlation-based networks to uncover how chronic conditions cluster together, revealing non-obvious comorbidity profiles across cardiac, metabolic, mental health, and other domains [5].

These network structures not only summarize existing disease associations but can also quantify directionality and timing in disease progression [6]. In practice, embedding comorbidity networks into predictive models allows a recommender system to account for a patient’s entire multimorbidity context: certain diseases may act as “hubs” or prerequisites for others, and network-based analytics can weight these dependencies when forecasting risk. Prior work has demonstrated that multimorbidity networks can improve clinical decision support, yielding better disease diagnosis and more holistic treatment recommendations, by reflecting how diseases interact within individuals [7]. Our recent effort used recurrent neural network-based approaches to leverage the diagnostic similarity and temporality of patients to improve prognostic modeling [8].

Critically, however, real-world clinical data are dynamic, and predictive models must adapt as new patient information arrives. Static, one-time-trained models quickly become outdated when population health or practice patterns shift — a phenomenon known as “calibration drift” [9, 10]. For instance, models trained on historical cohorts may underperform if disease prevalence or treatment protocols change. As populations and care evolve, static prediction models become outdated and less accurate over time. This limitation can lead to suboptimal or even harmful recommendations if left unaddressed. To counter this, dynamic predictive models are being considered: these continuously update their parameters in response to incoming data, ideally adjusting risk estimates on-the-fly as more observations accumulate. In machine learning terms, this is akin to *continual learning* or lifelong learning, where an algorithm incrementally learns from successive data batches while retaining prior knowledge [11]. In a healthcare recommender context, this might mean retraining or fine-tuning the model as each new patient’s data is recorded, rather than relying on a fixed model. Such continual adaptation is crucial in practice: without it, even sophisticated models can suffer from drift and lose generalizability across settings. In summary, while recommender-based predictive systems hold great promise for personalized healthcare, they must be designed with dynamic updating in mind. Only by continually integrating fresh patient data and avoiding the rigidity of static models can clinical recommender systems remain accurate and trustworthy in the face of evolving disease patterns and patient populations [12].

We envision a framework that uncovers clinically meaningful disease progressions beyond clinical disease classifications and refines them in near real-time from limited patient data (see Figure 1). We construct a directed disease progression network from co-occurrence data using the GENIE3 framework, enabling the capture of probabilistic influence relationships among diseases. The inferred network is subsequently sparsified via the Kneedle method to retain only the most informative edges while reducing noise and redundancy. Building on this trimmed structure, we propose an *incremental refinement algorithm* as the central methodological contribution, designed to mitigate the need for learning the network from scratch. This algorithm dynamically refines edge weights by assimilating streaming patient trajectories, updating disease influence relationships. Our theoretical analysis shows that the adjustment algorithm is a stable stochastic approximation: the bias in the estimate decays geometrically, ensuring asymptotic consistency with the true weight, while the variance remains bounded.

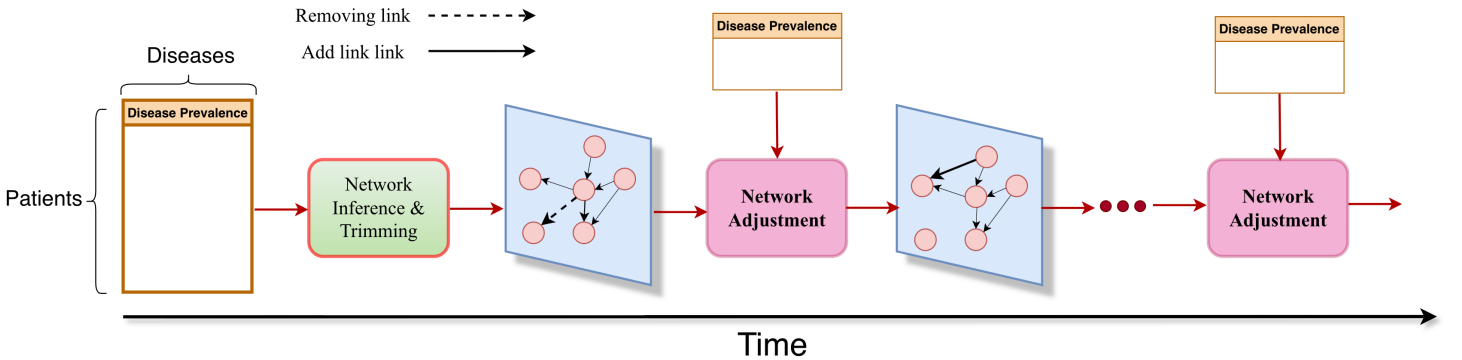


Figure 1: The framework performs network generation, trimming, followed by repeated adjustment to incoming patient data.

Analysis on the MIMIC-III dataset reveals that International Classification of Diseases, 9th Revision (ICD9) categories exhibit partial alignment with clusters inferred through co-occurrence structure, indicating both consistency with and deviations from traditional taxonomy. Our adjustment algorithm, when applied to the sparsified network, achieves strong agreement with the ground truth networks learned in near-linear time using the GENIE3-based approach as well as a baseline approach based on conditional probability, while learning from patient batches of size 50 within one second. Validation on clusters of quasi-patient trajectories sampled from the disease network demonstrates its ability to recover clinically meaningful disease progressions across diverse systems. These include obstetric complications cascading into respiratory failure, mucosal inflammation extending into gastrointestinal disorders, and gynecological conditions progressing toward pelvic inflammatory

disease. Complex cross-system dynamics are also captured, such as cardio-pulmonary coupling, neuro-obstetric-inflammatory interactions, and neurodegenerative pathways leading to cardiovascular complications. Lastly, the framework identifies psychiatric-reproductive linkages, post-stroke respiratory vulnerability, well-structured within-system progressions (e.g., prostate disorders), and chronic looping patterns in gastrointestinal disease, underscoring its capacity to uncover disease trajectories.

2 Approach

2.1 Preliminaries

2.1.1 Disease interaction represented as directed graphs

We consider a directed network $G(V, E)$, representing the interaction among diseases represented as nodes V . Each directed edge $(u, v) \in E$ is weighted, representing the influence of node u on v . The network inferred using a regression-based framework, GENIE3 [13]. For each node $u \in V$ representing a disease, its prevalence x_u is modeled as a function of all other nodes: $x_u = f(X_{-u}) + \epsilon$, where X_{-u} denotes the prevalence of all diseases except u , $f(\cdot)$ is a regression function capturing conditional dependencies, and ϵ is a noise term. The learned coefficients, interpreted as the *influence* of node u on v , serve as edge weights $w(u, v)$.

2.1.2 Influence diffusion

Diffusion describes the phenomenon of knowledge flow in networks with many applications [14, 15]. The seed nodes are considered *active* and initiate information spread to their downstream connections, who in turn influence their connections, and so forth. The influence of a node is measured by the number of downstream nodes it can activate. Given a set of seeds $S \subset V$ which are the initial active nodes, the diffusion proceeds as follows: an active node $u \in V$ is given a single chance to activate a neighbor v with probability given by its edge weight $w(u, v)$. If u is successful, v is activated at time $t + 1$ and acts as a spreader in the future. This continues until no further activations are possible, and the final activated count is recorded.

2.1.3 Clustering and network concordances

The concordance between clustering configurations (defined as known disease classification superclasses and modules identified from the inferred disease network) is the proportion of all disease pairs for which the inferred module membership agrees with the reference grouping. Analogously, we evaluate concordance accuracy between the ground-truth and inferred GENIE3 networks by comparing their edge weights. Specifically, we compute the Pearson correlation coefficient (PCC) to quantify linear agreement between corresponding edge weights, capturing how well the inferred magnitudes align with the true influence strengths. We use Kendall’s tau to assess rank concordance, measuring the consistency in the relative ordering of edge weights.

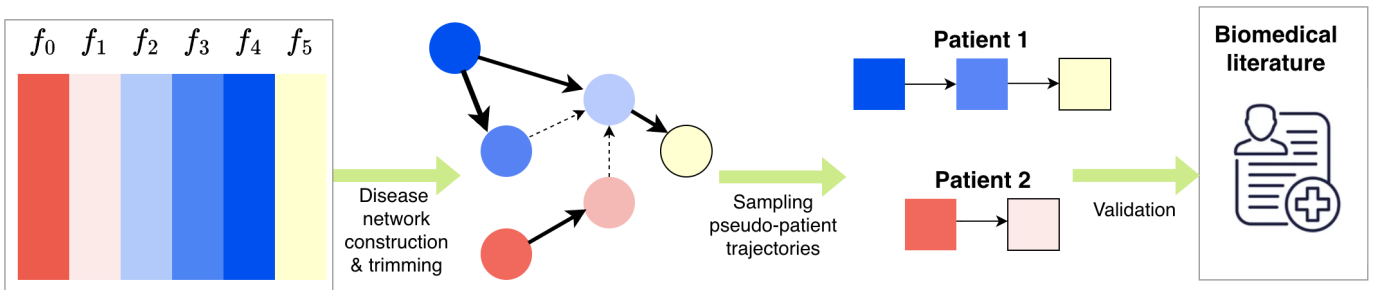


Figure 2: Validation of sparse disease network inferred from binary patient-disease data using sampled quasi-patient profiles.

2.2 Network Generation and Trimming

The biomedical dataset consists of patients as rows and binary disease prevalence features as columns, representing the presence or absence of each disease (see Figure 2). Using GENIE3, we infer a directed disease network where the edge weight $w_{u,v}$ quantifies the influence of disease u on disease v . The majority of the connections made by GENIE3 have a very low

measure of influence, with fewer high-weighted edges. To improve quality and interpretability, we approximate the “Knee point” for systematic edge trimming using the Kneedle method [16], which identifies the point of maximum curvature.

We iteratively remove edges with the smallest weights from the complete directed disease network. Given the total number of edges E_0 , at each step t , we remove the edge e_t with the minimal weight $w_{\min}$ and compute the fraction of remaining edges: $f(t) = \frac{|E_0| - t}{|E_0|}$. Let $f(t)$ be normalized to $[0, 1]$ as $\tilde{f}(t)$, and define the difference function: $d(t) = \tilde{f}(t) - t/T$, where T is the total number of pruning steps. The knee point t^* corresponds to the step where $d(t)$ attains its maximum: $t^* = \arg \max_t d(t)$.

Algorithm 1 Disease network update through incremental refinement with patient trajectories

0: **Input:** Current directed weighted network G_0 , number of patients H , step size δ , decay rate γ
0: **Output:** Evolved network G

```

0: Initialize  $G \leftarrow G_0$ 
0: Initialize  $\delta_0 \leftarrow \delta$ 

0: while not converged (or for  $T$  iterations) do
0:   1. Construct Edge Frequency Subnetwork  $H$ :
0:   Initialize  $H \leftarrow \emptyset$ , edge frequency map  $f(\cdot, \cdot) \leftarrow 0$ 
0:   for  $h = 1$  to  $H$  do
0:     Sample seed set  $S_h$  from nodes in  $G_0$ 
0:     Generate trajectory sequence  $Seq_h \sim \text{spread\_reach\_chains}(G_0, S_h, L)$ 
0:     for each sequence  $seq \in Seq_h$  do
0:       for  $i = 1$  to  $|seq| - 1$  do
0:          $f(seq_i, seq_{i+1}) \leftarrow f(seq_i, seq_{i+1}) + \frac{1}{H}$ 
0:       end for
0:     end for
0:   end for
0:   Construct  $H$  with edges  $(u, v)$  weighted by  $f(u, v)$ 

0:   2. Store Total Weight:
0:    $W_{\text{old}} \leftarrow \sum_{(u,v) \in G} w_G(u, v)$ 

0:   3. Update Informative Edges (in  $H$ ):
0:   for each  $(u, v)$  with  $f(u, v) > 0$  do
0:     if  $(u, v) \notin G$  then
0:       Add edge  $(u, v)$  with  $w_G(u, v) \leftarrow 0$ 
0:     end if
0:      $w_G(u, v) \leftarrow (1 - \delta) w_G(u, v) + \delta f(u, v)$ 
0:      $w_G(u, v) \leftarrow \min(1, \max(0, w_G(u, v)))$ 
0:   end for

0:   4. Compute Weight Drift:
0:    $W_{\text{new}} \leftarrow \sum_{(u,v) \in G} w_G(u, v)$ 
0:    $\Delta \leftarrow W_{\text{new}} - W_{\text{old}}$ 

0:   5. Conservation Redistribution over Non- $H$  Edges:
0:    $L \leftarrow \{(u, v) \in G : f(u, v) = 0\}$ 
0:    $S \leftarrow \sum_{(u,v) \in L} w_G(u, v)$ 
0:   if  $S > 0$  then
0:     for each  $(u, v) \in L$  do
0:        $w_G(u, v) \leftarrow w_G(u, v) - \frac{w_G(u, v)}{S} \cdot \Delta$ 
0:        $w_G(u, v) \leftarrow \min(1, \max(0, w_G(u, v)))$ 
0:     end for
0:   end if

0:    $\delta \leftarrow \gamma \cdot \delta$ 
0: end while
0: return  $G = 0$ 

```

2.3 Incremental Refinement of the Disease Network

Algorithm 1 maintains a trimmed (or sparsified) directed weighted disease influence graph G , where edge weights encode transition or activation strengths inferred from patient trajectories. At each iteration, a batch of H patients is sampled, and for each patient, disease progression is simulated as diffusion trajectories over a fixed horizon. A trajectory is decomposed into ordered node pairs, and the frequency of each observed edge (u, v) is accumulated to construct an empirical edge-frequency subnetwork H , where $w_H(u, v)$ represents the normalized frequency of occurrence of edge (u, v) across the patient batch.

The current network G is then updated using this empirical evidence. For every edge observed in H , its weight in G is updated via a convex combination, $w_G(u, v) \leftarrow (1 - \delta) w_G(u, v) + \delta w_H(u, v)$, and edges not already present in G are

first introduced with zero weight before updating. All updated weights are projected onto $[0, 1]$ to maintain stability. This update induces a global change in total edge mass, quantified by Δ , which is computed as the difference between the post-update and pre-update total weight of the network. To enforce conservation of total interaction mass, this excess or deficit is redistributed over edges not supported by the observed subnetwork H . Specifically, weights of non-observed edges are adjusted proportionally to their current magnitudes, $w_G(u, v) \leftarrow w_G(u, v) - \frac{w_G(u, v)}{\sum_{(x, y): w_H(x, y)=0} w_G(x, y)} \cdot \Delta$, followed again by projection onto $[0, 1]$. This mechanism reinforces edges supported by empirical patient flows while systematically attenuating unsupported interactions, resulting in a network that evolves consistently with observed diffusion dynamics.

Algorithm 2 Quasi-patient trajectory generation and disease subtype discovery

Require: Disease graph $G(V, E, W)$, number of patients H , trajectory length l , cluster range K

0: **Load disease network** G with weighted edges

0: Normalize edge weights: $W_{uv} \leftarrow \frac{W_{uv}}{\max(W)}$

0: Initialize empty list of patient trajectories: $\mathcal{S} \leftarrow \emptyset$

0: **for** each patient $h = 1 \dots H$ **do**

0: Sample initial seed disease s using degree-weighted distribution

0: Generate trajectory $T_h \leftarrow \text{WeightedRandomWalk}(G, s, l)$

0: Append T_h to $\mathcal{S}$

0: **end for**

0: **function** $\text{WEIGHTEDRANDOMWALK}(G, s, l)$

0: Initialize path $P \leftarrow [s]$

0: **while** $|P| < l$ **do**

0: Let current node be u

0: Sample next node $v \sim P(v|u)$ proportional to edge weights

0: Append v to P

0: **end while**

0: **return** P

0: **end function**

0: Compute pairwise trajectory distance matrix D

0: **for** each pair (T_i, T_j) **do**

0: Extract contiguous edge sets E_i, E_j

0: $D_{ij} \leftarrow \frac{1}{|E_i \cap E_j| + 1}$

0: **end for**

0: **Cluster trajectories**

0: **for** $k \in K$ **do**

0: Perform Agglomerative Clustering on D

0: Compute silhouette score s_k

0: **end for**

0: Select optimal clustering $k^* = \arg \max_k s_k$

0: **Temporal motif extraction**

0: **for** each cluster c **do**

0: Extract trajectories $\mathcal{S}_c$

0: Compute edge-frequency tensor over time:

$D(u, v, t) = \text{frequency of transition } u \rightarrow v \text{ at step } t$

0: Aggregate dominant temporal transitions per cluster

0: **end for**

0: **Output:**

0: Disease subtypes (clusters of trajectories)

0: Temporal progression motifs per subtype

0: =0

Remark 1. The expected weight w in Algorithm 1 converges to the true value p in the long run.

Proof. As per the update rule, $w_{t+1} = (1 - \delta)w_t + \delta\hat{p}$, where w_t is the time-varying update of the estimated GENIE weight and $\hat{p}$ is the new empirical measure.

$$\implies \mathbf{E}[w_{t+1}] = (1 - \delta)\mathbf{E}[w_t] + \delta\mathbf{E}[\hat{p}].$$

Let $\mu_t = \mathbf{E}[w_t]$ and $\mathbf{E}[w_t] = p$. Then, $\mu_{t+1} = (1 - \delta)\mu_t + \delta p$.

$$\implies \epsilon_{t+1} = \mu_{t+1} - p = (1 - \delta)\mu_t + \delta p - p.$$

$$\implies \epsilon_{t+1} = (1 - \delta)(\mu_t - p) = (1 - \delta)\epsilon_t.$$

Recursively, $\epsilon_{t+1} = (1 - \delta)^t \epsilon_0$. As, $t \rightarrow \infty$, $\epsilon_{t+1} \rightarrow 0$.

□

Remark 2. The prediction error in Algorithm 1 oscillates around 0 with a variance of $\frac{\delta}{2-\delta} \times \sigma_\eta^2$.

Proof. $\text{VAR}[e_{t+1}] = (1 - \delta)^2 \text{VAR}[e_t] + \delta^2 \sigma_n^2 \implies v_{t+1} = (1 - \delta)^2 v_t + \delta^2 \sigma_n^2$.
 $\implies v_{t+1} = (1 - \delta)^{2t} v_0 + \delta^2 \sigma_n^2 \sum_{k=0}^{t-1} (1 - \delta)^{2k}$
 $\implies v_{t+1} = (1 - \delta)^{2t} v_0 + \delta^2 \sigma_n^2 \frac{1 - (1 - \delta)^{2t}}{1 - (1 - \delta)^2}$

As $t \rightarrow \infty$, the first term $\rightarrow 0$ and the second term $\rightarrow \frac{\delta}{2 - \delta} \sigma_n^2$ □

Remark 3. Under sparse network assumptions where $|E| = O(|V|)$, and with localized diffusion under the Independent Cascade model, the per-iteration complexity of Algorithm 1 is linear in the number of nodes.

Proof. In Step 1, each of the H patient trajectories is generated via an Independent Cascade process, where each edge is attempted once. Under sparse and localized diffusion, the number of explored edges per cascade is $O(|V|)$ in expectation, yielding a total cost of $O(H \cdot |V|)$. Steps 2–5 involve computing total weight, updating edges, and redistributing weights, each requiring a traversal over the edge set, contributing $O(|E|) = O(|V|)$. Combining these, the per-iteration complexity is $O(H \cdot |V| + |V|) = O(H \cdot |V|)$. For fixed batch size H , this reduces to $O(|V|)$, establishing linear-time complexity. □

Collectively, we make these remarks to demonstrate that the update rule is both *consistent* and *stable*: the estimated weights converge in expectation to the true underlying values while the stochastic fluctuations around this limit remain bounded. In particular, the parameter δ governs a bias–variance trade-off, where larger values accelerate convergence but induce higher steady-state variability, and smaller values yield smoother, stable estimates at the cost of slower adaptation.

ICD9 Code Range	Superclass
1–139	Infectious and Parasitic Diseases
140–239	Neoplasms
240–279	Endocrine, Nutritional and Metabolic Diseases, and Immunity Disorders
280–289	Diseases of the Blood and Blood-forming Organs
290–319	Mental Disorders
320–389	Diseases of the Nervous System and Sense Organs
390–459	Diseases of the Circulatory System
460–519	Diseases of the Respiratory System
520–579	Diseases of the Digestive System
580–629	Diseases of the Genitourinary System
630–679	Complications of Pregnancy, Childbirth, and the Puerperium
680–709	Diseases of the Skin and Subcutaneous Tissue
710–739	Diseases of the Musculoskeletal System and Connective Tissue
740–759	Congenital Anomalies
760–779	Certain Conditions originating in the Perinatal Period
780–799	Symptoms, Signs and Ill-defined Conditions
800–999	Injury and Poisoning

Table 1: ICD9 code ranges and corresponding disease superclasses.

2.4 Validation on Quasi-patient Trajectories

The validation, discussed in Algorithm 2, once again begins by taking a weighted disease network $G(V, E, W)$, where nodes are diseases and edge weights encode transition likelihoods between diseases. All edge weights are first normalized by the maximum weight to ensure comparability across transitions. The algorithm then generates a collection of quasi-patient trajectories by simulating stochastic disease progression on this network (see Figure 2). For each of H patients, an initial disease is sampled using a degree-weighted scheme, and a fixed-length trajectory is produced via a weighted random walk, where each next disease is drawn proportionally to the outgoing edge weights of the current node. This process yields a population-level dataset of synthetic disease sequences representing plausible temporal disease progressions. Once trajectories are generated, the method constructs a similarity structure between patients by defining a distance metric based on overlapping contiguous disease transitions. Specifically, each trajectory is decomposed into ordered disease pairs, and pairwise distances are computed as the inverse of the number of shared transitions between trajectories. This distance matrix is then used as input to hierarchical agglomerative clustering, and the number of clusters is selected by maximizing a validation criterion such as the silhouette score [17] over a range of candidate cluster sizes. Finally, within each discovered cluster, temporal transition frequencies are computed by aggregating edge occurrences across trajectory positions, producing a time-resolved transition tensor. These cluster-specific temporal patterns reveal distinct disease subtypes characterized not only by shared diseases but by shared progression dynamics, enabling interpretation of clinically meaningful temporal motifs in disease evolution.

Parameter	Value
GENIE	
Number of trees	200
Learning model	Random Forest
Adjustment Algorithm	
Integration of new data (δ)	0.05
Decay factor (γ)	0.9985
Length of sequence	25
Number of new patients	50
Number of seeds	5
Duration of influence diffusion	1000

Table 2: Default parameter settings.

3 Results

We conducted the analysis using the MIMIC-III dataset [18], a de-identified health database of patients admitted to critical care units at a tertiary care hospital from 2001 to 2012. The dataset includes demographics, vital signs, hospital admissions, laboratory test results, procedures, medications, and diagnosis codes. It employs ICD9 codes for recording diagnoses and procedures, a system for categorizing health data (see Table 1 for the ICD9 code ranges and superclass descriptions).

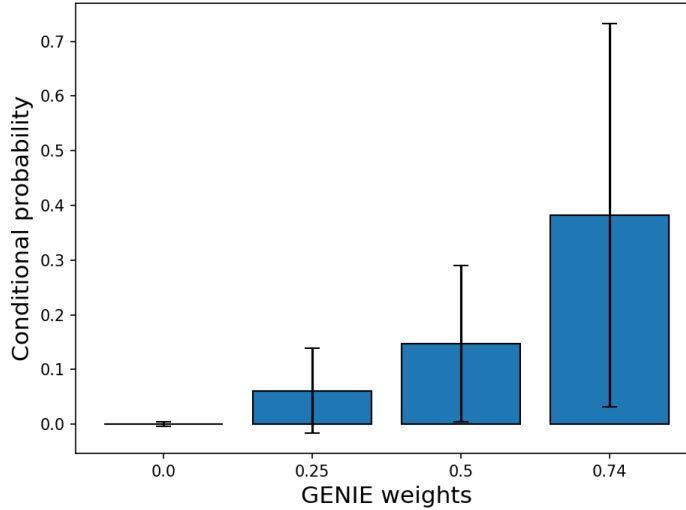


Figure 3: Conditional probability of disease prevalence as a function of GENIE weights (mean $\pm$ SD).

All experiments were implemented in Python, leveraging a combination of established libraries for network construction, inference, and analysis. Graph operations, including creation, traversal, and dynamic updates, were performed using **NetworkX** [19]. The underlying disease interaction network was inferred using the GENIE3 framework [13], which enables tree-based ensemble learning for reconstructing weighted directed graphs from high-dimensional data. To ensure tractability and interpretability, the inferred network was sparsified using the Kneedle method [16] to identify an appropriate threshold in the edge-weight distribution. For visualization and exploratory analysis, the resulting networks were exported and examined in **Cytoscape** [20], which provides flexible tools for inspecting structural patterns and edge-weight distributions. Unless otherwise specified, all model and algorithmic parameters were set to their default values, which are enumerated in Table 2.

3.1 Preprocessing and Generation of the Disease Network

As a preprocessing step, we removed all disease columns with no observed occurrences (i.e., columns containing only zeros or NaN values) from the MIMIC-III dataset, ensuring that the retained matrix reflects meaningful clinical activity and avoids

degenerate features. The resulting matrix was then used as input to the GENIE framework, which was executed in random forest mode with 200 trees. The inferred GENIE network is a simple, complete, directed, and edge-weighted graph comprising 6,985 disease nodes and 48,783,240 edges, where edge weights represent learned dependencies between disease pairs.

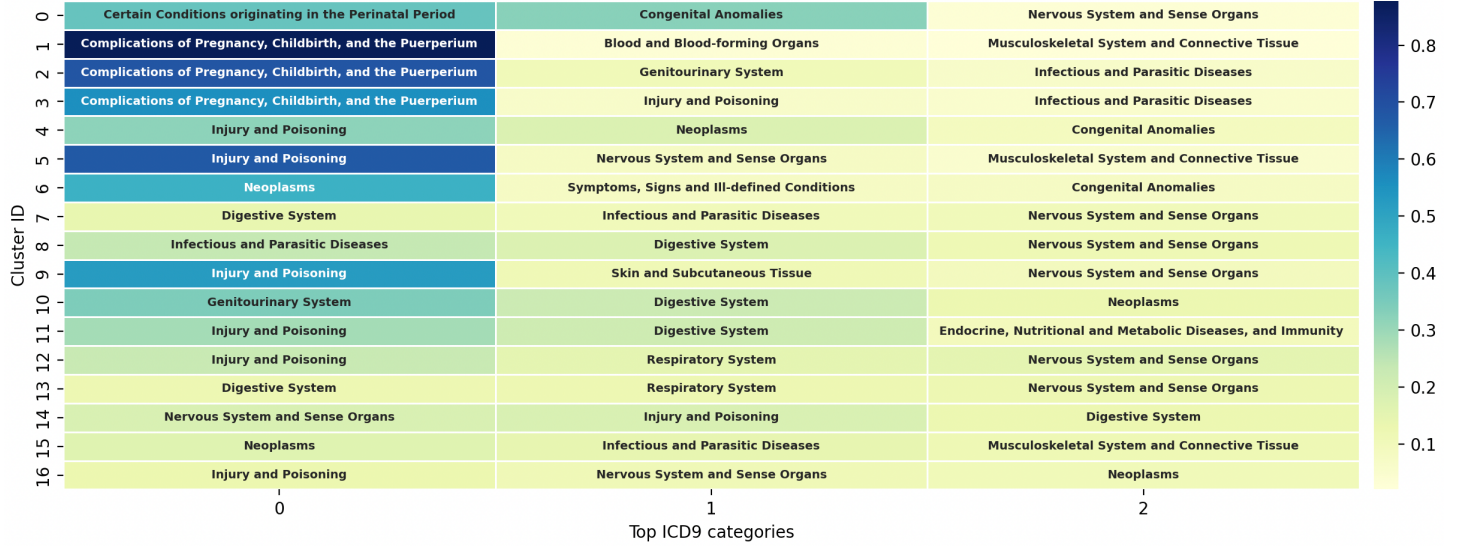


Figure 4: Distribution of the top-3 ICD9 tags within the 17 GENIE3-based disease network

3.2 Relationship between Influence Weights and Conditional Prevalence

To assess whether the learned GENIE edge weights $w(u, v)$ reflect meaningful prevalence relationships, we performed a stratified analysis based on the magnitude of these weights. Specifically, we partitioned all edges into four groups (quartiles) according to their corresponding $w(u, v)$ values. For each group, we computed the conditional probability $w(u, v)$, defined as the empirical probability of observing node v given the presence of node u across the clinical dataset, i.e., $\frac{\#(u \cap v)}{\#(u)}$. Within each weight group, we then calculated the mean and standard deviation of $w(u, v)$. Figure 3 presents the mean $\pm$ one standard deviation of $w(u, v)$ across the four weight groups. We observe a clear positive trend: edges with higher GENIE weights correspond to higher conditional probabilities. This suggests that the GENIE weights capture underlying prevalence patterns in the data, with stronger weights indicating stronger associative or co-occurrence relationships between nodes.

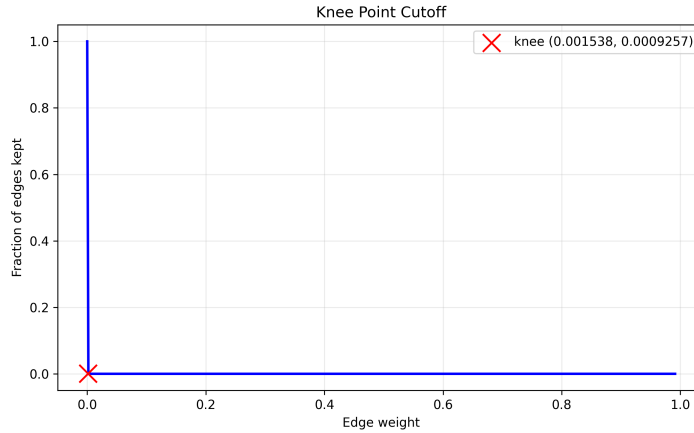


Figure 5: Identification of knee point for disease network sparsification.

Given that the disease space is organized into 17 ICD9 superclasses, we construct a 17-cluster representation in the GENIE-based disease network. To achieve this, we apply iterative hierarchical clustering [21] combined with a pruning strategy that removes low-support disease nodes, ensuring that each resulting cluster contains at least 10 diseases. This iterative refinement continues until cluster stability is reached. Following this, we quantify the distribution of ICD9 disease codes within each GENIE module to characterize its clinical composition. In addition, we evaluate clustering concordance using a pairwise *accuracy* metric, defined as the proportion of all disease pairs that are either co-assigned to the same GENIE module and the same ICD9 superclass, or simultaneously assigned to different GENIE modules and ICD9 superclasses.

Figure 4 shows that the GENIE-based clusters are heterogeneous, containing mixtures of multiple ICD9 disease superclasses, indicating that prevalence-driven network structure does not directly align with clinical taxonomy. This mismatch is reflected in a moderate concordance of approximately 0.59, suggesting partial agreement between inferred network modules and ICD9-based clinical subtyping (see Section 2.1.3). Despite this heterogeneity, the top three ICD9 classes within each prevalence-based module reveal interpretable co-occurrence. Specifically, one dominant pattern links pregnancy-related conditions with genitourinary and infectious diseases, reflecting maternal-fetal and perinatal complication pathways where infection and urogenital disorders act as key comorbid drivers [22]. A second consistent module connects injury and poisoning with nervous system and musculoskeletal diseases, forming a trauma-neurological-rehabilitation axis that likely reflects cascades from physical injury to neurological sequelae and multi-system involvement [23]. A third pattern groups neoplasms with congenital anomalies and infectious diseases, suggesting a comorbidity structure associated with immunocompromised states and systemic disease [24]. Finally, a metabolic-organ system coupling module emerges, linking digestive, genitourinary, and endocrine disorders, consistent with chronic disease progression and systemic metabolic dysregulation patterns [25].

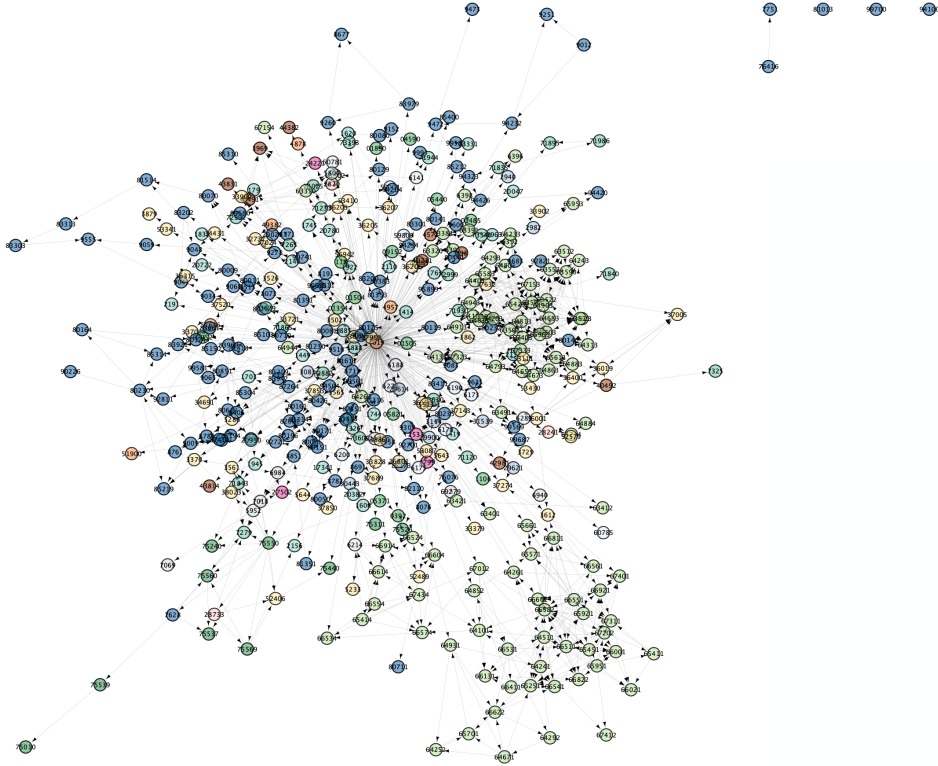
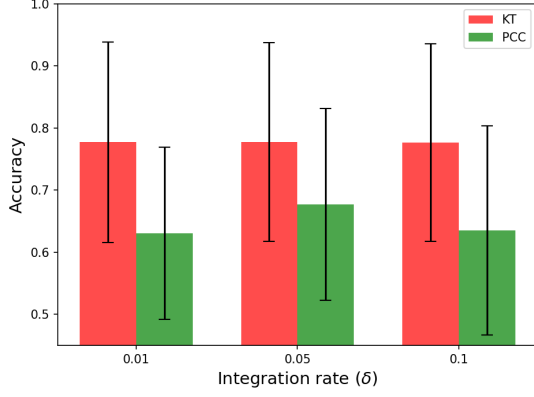
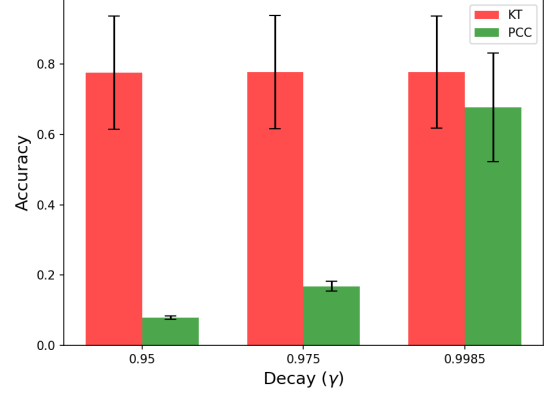


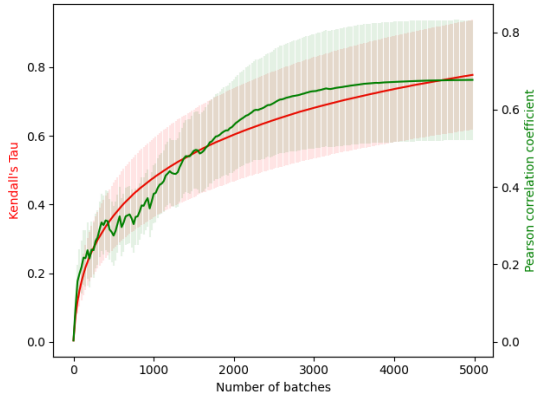
Figure 6: A 500-node subnetwork extracted by performing depth-first search on the trimmed GENIE3-based disease network, where diseases are color-coded by their ICD9 superclasses.



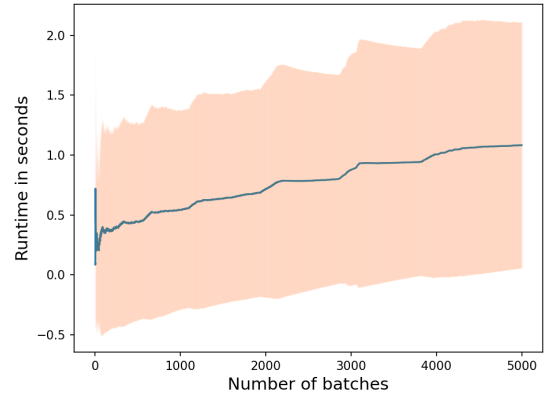
(a) Constant decay $\gamma = 0.05$



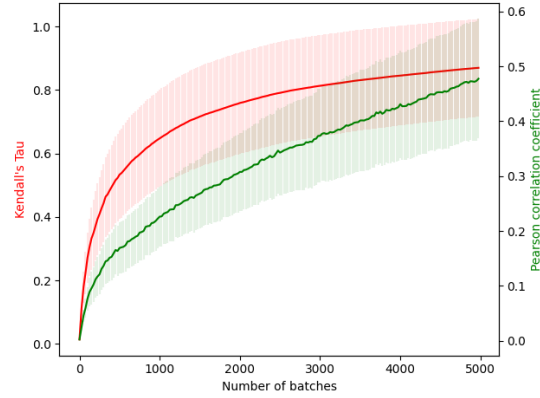
(b) Constant integration rate $\delta = 0.985$



(c) $\delta = 0.05, \gamma = 0.9985$



(d) $\delta = 0.05, \gamma = 0.9985$



(e) $\delta = 0.3, \gamma = 0.9999$

Figure 7: Iterative refinement of the inferred disease network initialized as a sparse random network over 5000 batches: Variation in concordance for changing (a) integration rate (δ); and (b) decay in integration rate (γ); (c) mean concordance ± 1 std. between GENIE-based ground truth and inferred disease networks; (d) mean runtime ± 1 std. (per batch) of the adjustment algorithm in seconds; and (e) mean concordance for the conditional probability-based disease network.

3.3 Disease Network Sparsification

The GENIE-based disease network was sparsified using the Kneedle method to identify an appropriate threshold in the edge-weight distribution. This procedure revealed a clear knee point at a GENIE weight of 0.0015, which was used as the cutoff for pruning weak interactions. Accordingly, all edges with weights below 0.0015 were removed from the network, resulting in a significantly sparser and more stable graph that preserves the dominant disease-transition structure (see Figure 5). All subsequent experiments in this study were conducted on this sparsified network of 6985 nodes and 680370 directed edges.

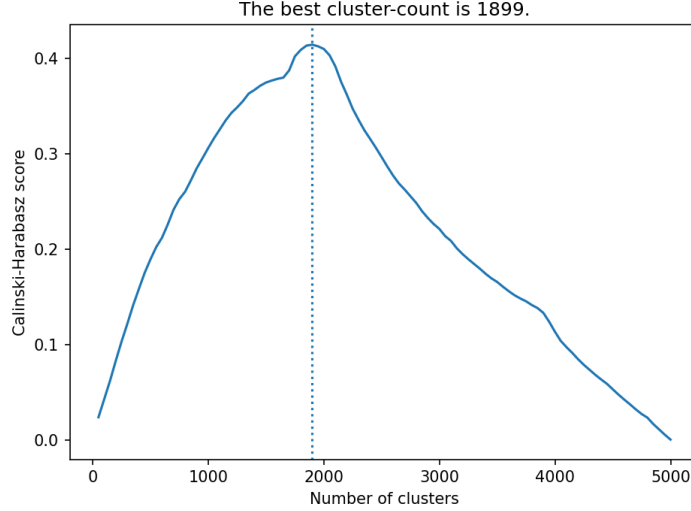


Figure 8: Identification of the patient cluster configuration using the Silhouette Score

3.4 Concordance between Prevalence and Clinical Classification

We extract a representative 500-node subnetwork of the trimmed GENIE-based disease network by performing a depth-first search (DFS) starting randomly from a disease node selected with a probability proportional to its degree, yielding a connected subgraph of 500 nodes that preserves locally dense, high-influence regions of the network. This DFS-based sampling prioritizes traversal along dominant outgoing pathways, capturing coherent progression patterns embedded in the directed structure. The resulting subnetwork is visualized in Cytoscape, where nodes are color-coded according to their ICD9 superclasses (refer to Figure 6). The dispersed mix of node colors corroborates the partial concordance of 0.59 discussed in Section 3.2, indicating that disease prevalence patterns do not strictly align with ICD9 superclass boundaries.

3.5 Incremental Refinement of Disease Network

We experimented with 5,000 iterative batches, where a batch incorporated 50 synthetic (or “pseudo”) patients, each represented by a disease progression sequence of length 25 and generated via random walks on an unobserved trimmed ground-truth disease network. The inferred network was initialized as a sparse, random directed graph and iteratively updated using the observed transition frequencies from patient trajectories. To assess convergence toward the ground truth, we tracked both Kendall’s tau (KT) and the Pearson correlation coefficient (PCC) between the inferred and true edge weight over time.

Figures 7(a) and 7(b) show the effect of varying the integration rate (δ) and decay (γ) parameters, keeping the other parameter constant at the default value. Across all settings, KT remains consistently high (0.77 with similar variance), indicating that the rank ordering of edge weights in the learned network is robust to the initial update strength (δ) and its decay. In contrast, PCC, which is sensitive to absolute magnitudes, varies with the decay parameter. When decay is slow (e.g., 0.9985), δ persists longer, allowing cumulative updates to better match the true weight scale, yielding moderate PCC (0.63–0.68). However, with aggressive decay (0.95 or 0.975), δ shrinks rapidly, freezing the network before sufficient magnitude correction occurs—this preserves relative ordering (high KT) but collapses absolute agreement (very low PCC,

0.08–0.17). Therefore, δ primarily controls update intensity, while decay governs the temporal window of learning, with PCC revealing sensitivity to outliers and insufficient long-term adaptation, whereas KT reflects stable structural recovery.

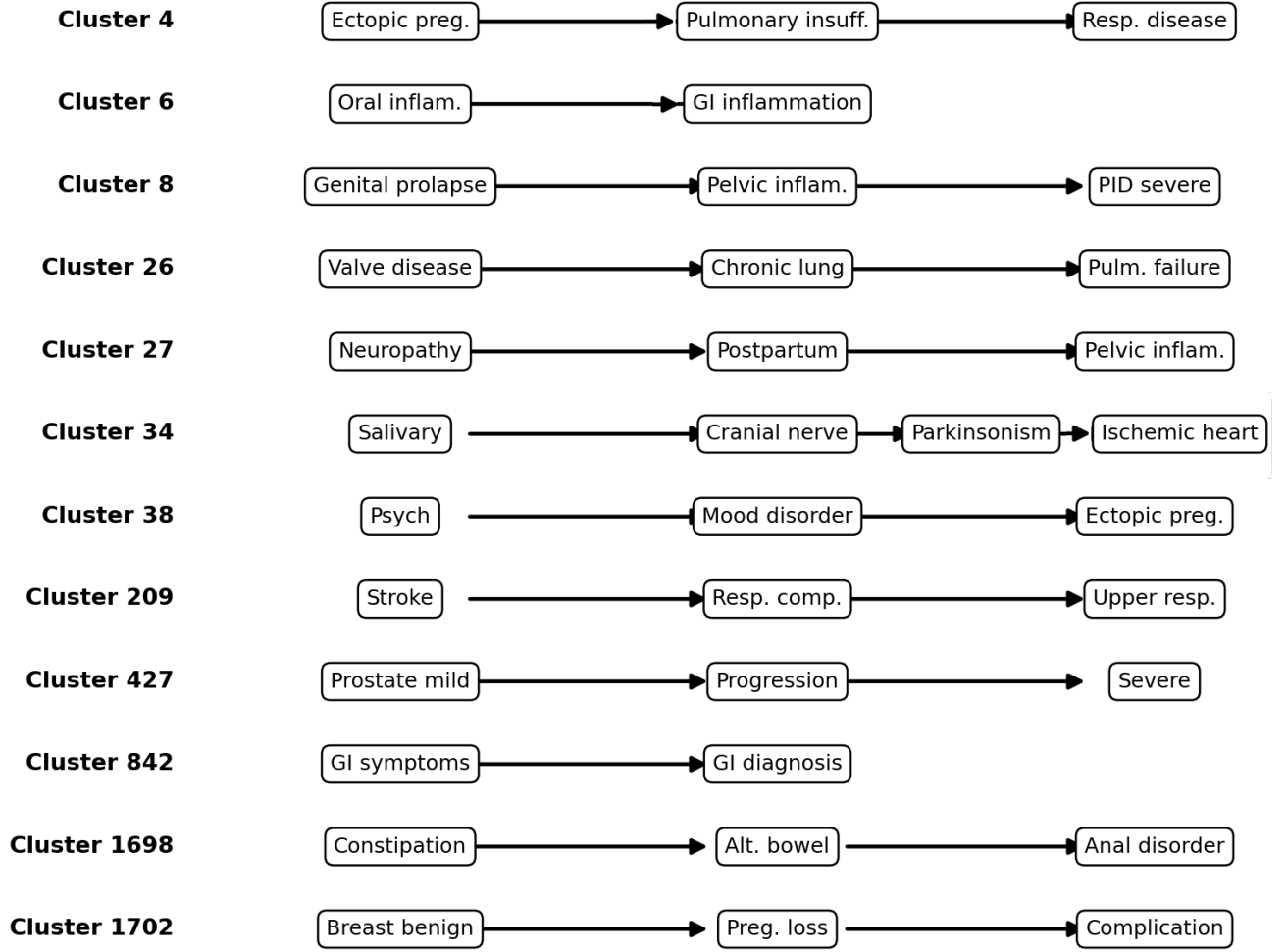


Figure 9: Disease profiles and temporal disease progression trends within patient clusters.

Figure 7(c) shows the results from the default configuration, given by $\delta = 0.05$ and $\gamma = 0.985$. We observe that as the number of batches increased, both metrics exhibited steady improvement, reaching values of 0.77 and 0.67 for KT and PCC, respectively, indicating recovery of the underlying network structure. In addition to accuracy, we evaluated the computational efficiency of the update mechanism. Despite the progressive densification of the inferred network, the runtime per batch, each involving 50 patient trajectories, remained bounded under 2 seconds (avg. runtime is 1.08 ± 1.02 s) on a standard machine with 16 GB of memory. This is orders of magnitude faster than running GENIE3 from scratch, where each pass of GENIE3 on 5000 patients took 3158 s. Figure 7(d) shows that the method scales with accumulating data, making it suitable for iterative or streaming settings where trajectories are incorporated sequentially. Lastly, a conditional-probability baseline [26], following the same formula for measuring influence of disease u on v , $w(u, v)$ applied in Section 3.2, yields a KT of approximately 0.9 and a PCC of approximately 0.5 (see Figure 7(e)), indicating that the adjustment algorithm preserves a strong monotonic ordering between true and inferred influence weights, though the relationship is not strictly linear.

3.6 Case Studies on Patient Trajectories

To evaluate population-level structure, we follow an approach similar to [27] to simulate a cohort of 5,000 quasi-patients, in which each patient’s disease trajectory is generated as a sequence of ICD-9 codes via random walks on the sparsified disease

network. These trajectories capture plausible temporal co-occurrence patterns induced by the learned influence. We then perform clustering over the resulting patient representations, selecting the optimal number of clusters based on the Silhouette score to ensure well-separated groupings (see Figure 8). Finally, the patient clusters (some of which are visualized in Figure 9) are examined to uncover meaningful patterns of disease evolution, enabling interpretation of characteristic progression pathways.

Temporal disease trajectories across clusters. We observe that several clusters exhibit coherent and clinically interpretable temporal progressions. In Cluster 4, the trajectory $6332 \rightarrow 5152 \rightarrow 4858$ links ectopic pregnancy to downstream pulmonary insufficiency, suggesting severe obstetric events cascading into systemic inflammatory or respiratory compromise [28]. Cluster 6 ($5283 \rightarrow 5586$) reflects a mucosal-immune axis, where oral inflammatory conditions precede gastrointestinal inflammation [29]. Similarly, Cluster 8 ($6187 \rightarrow 6147 \rightarrow 6146$) captures a gynecological progression from structural abnormalities (genital prolapse) to pelvic inflammatory disease [30]. In Cluster 26 ($3979 \rightarrow 4995 \rightarrow 4994$), a cardio-pulmonary coupling emerges, consistent with the interdependence between valvular heart disease and chronic pulmonary dysfunction [31].

Cross-system interactions and systemic decline. Cross-system interactions appear in Cluster 27 ($3550 \rightarrow 6781 \rightarrow 6145$), where neurological, obstetric, and inflammatory processes co-evolve, reflecting systemic stress or immune dysregulation during the peripartum period. Cluster 34 ($5278 \rightarrow 3505 \rightarrow 3324 \rightarrow 4146$) suggests a progression from peripheral dysfunction to neurodegenerative disease and eventual cardiovascular complications, indicative of aging-related comorbidity accumulation [32]. In Cluster 38, trajectories involving $2974 \rightarrow 2964 \rightarrow 6339$ point to psychiatric conditions preceding reproductive complications, highlighting a mental health-reproductive health coupling [33]. Cluster 209 ($4340 \rightarrow 4690 \rightarrow 4705$) captures post-stroke respiratory vulnerability, consistent with aspiration risk and infection following neurological insult [34].

Within-system progression and chronic loops. Other clusters show localized or cyclic disease dynamics. Cluster 427 ($6003 \rightarrow 6004 \rightarrow 6002$) shows a clean progression within prostate disorders, likely reflecting worsening severity or diagnostic refinement [35]. Cluster 842 ($538x \rightarrow 5373$) represents the consolidation of nonspecific gastrointestinal symptoms into a defined diagnosis. Cluster 1698 ($5647 \leftrightarrow 5648 \leftrightarrow 5651$) exhibits a chronic looping structure, characteristic of relapsing functional gastrointestinal disorders [36]. Finally, Cluster 1702 ($2170 \rightarrow 6383 \rightarrow 6367$) links benign breast disease to adverse pregnancy outcomes, suggesting underlying hormonal or systemic predispositions influencing reproductive health trajectories [37].

4 Discussion

Our work emphasizes the need to move beyond static disease inference models toward adaptive frameworks capable of supporting meaningful temporal prognostication in healthcare recommender systems. The motivation is that clinical environments are inherently non-stationary: disease prevalence shifts, treatment protocols evolve, and patient populations differ across institutions and time. In this setting, models that require repeated retraining become computationally impractical and conceptually misaligned with real-world deployment constraints under concept drift. Our framework addresses this gap by enabling incremental refinement of a learned disease progression network, allowing it to evolve in near-linear time in response to incoming patient trajectories without reconstructing the underlying inference model from scratch. The key methodological contribution is an adjustment mechanism that updates edge weights through streaming data assimilation, coupled with theoretical guarantees showing that the resulting stochastic approximation is stable, with geometrically decaying bias and bounded steady-state variance. Empirically, this leads to strong agreement with full re-estimation procedures while maintaining high computational efficiency.

Future extensions must account for the frequency and persistence of diagnoses, rather than treating all observed transitions uniformly. In practical clinical environments, repeatedly occurring disease progressions may carry different prognostic significance compared to infrequent or isolated transitions, motivating the incorporation of diagnosis-frequency-aware weighting schemes during network refinement. Real-world deployment additionally requires adapting the update process to institution-specific operational constraints, since hospitals differ in patient throughput, reporting cadence, and data availability. This introduces important logistical tradeoffs involving the minimum number of patient trajectories required before triggering reliable edge-weight updates, as well as determining how frequently synchronization and refinement should occur to satisfy accuracy and stability requirements. Balancing these updated policies against computational and communication costs will

be essential for ensuring scalable deployment without imposing unnecessary infrastructure burden on participating health-care centers. Other caveats remain when moving toward real-world multi-center deployment. In decentralized collaborative environments, heterogeneity across institutions can induce conflicting updates, where locally consistent trajectory signals may not align globally. This raises challenges related to update synchronization, bias amplification across sites, and the potential for over-reinforcement of center-specific artifacts. In such cases, selective attention mechanisms become necessary to prioritize reliable transitions while down-weighting spurious or institution-specific patterns.

A further limitation arises in determining when the system should *unlearn* outdated edges. Our adjustment algorithm takes care of this by removing influence weights that drop below 0 (see Step 5 in Algorithm 1). While incremental updates provide adaptability, they may also accumulate historical bias if outdated clinical regimes persist in the network structure. This motivates the need for a predictive control model that can identify when structural revisions or partial unlearning should be triggered, under regime shifts or detected distributional drift. Such a mechanism would complement the current update rule by introducing a higher-level decision layer governing when adaptation should be local (weight refinement) versus structural (edge suppression or re-initialization). In a multi-center setting, these challenges become more pronounced. Institutions may differ in coding practices, patient demographics, and care pathways, leading to non-identical observational distributions. Extending the framework to federated or peer-to-peer learning settings would require handling of asynchronous updates, communication constraints, and privacy-preserving aggregation of trajectory information. Despite these challenges, this framework provides a foundation for continuously evolving clinical knowledge graphs that remain computationally efficient while being responsive to real-time data. In the long term, this work suggests a shift toward adaptive healthcare recommender systems that function as living representations of disease dynamics. Future directions include incorporating uncertainty-aware update gating, integrating causal constraints to improve interpretability, and developing explicit drift-detection modules to guide when adaptation should occur. These extensions are essential for robustness in decentralized clinical environments where data and underlying disease processes are in constant flux.

Competing Interests Statement

The authors declare no competing interests.

Data Availability Statement

The Python software code used to implement and evaluate the proposed framework is available from the corresponding author upon reasonable request and will be publicly released at <https://github.com/satunr/IterativeRefinement>.

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